

Riboflavin and the risk of dementia

A Healthy Skepticism Case Study

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The claim

On 29 August 2026 EatingWell ran an article under the headline "This Vitamin May Cut Dementia Risk Almost in Half, Research Says." The vitamin is riboflavin, vitamin B2. The research is one Japanese study of 4,171 adults. The article says people who ate the most riboflavin had "up to a 49% lower risk" of disabling dementia than people who ate the least, and it ends by recommending dairy, eggs, fish, and leafy greens.

The article is better than most. It says the study cannot prove cause and effect and it lists the study's limitations. The headline still promises something the study did not find. The article makes three claims and a recommendation, and each carries a fault. This paper checks each one against the study's own tables.

Manning J. EatingWell, 29 Aug 2026. Kishida R, Yamagishi K, Maruyama K, et al. Eur J Clin Nutr 2025;79:1099–1105. The published paper is paywalled; the counts below come from the authors' preprint of the same study (Research Square, 20 Feb 2024, doi 10.21203/rs.3.rs-3874560/v1), which sorts intake into fifths.

What the study did

Between 1985 and 1999, dietitians in three Japanese communities asked 4,171 adults aged 40 to 69 what they had eaten in the previous 24 hours. One day of food, asked once. The researchers estimated each person's riboflavin intake, sorted people into fifths, and then watched Japan's long-term care insurance records for anyone certified as needing daily help because of dementia. Over a median of 15.4 years, 887 of the 4,171 were certified, about 21 in 100. The records do not say what kind of dementia it was.

Claim 1. The vitamin may cut dementia risk almost in half

The fault: relative reporting. A percentage is reported without the counts it came from.

The 49% is a hazard ratio of 0.51, the output of a statistical model. The counts are in the preprint's Table 2. In the lowest fifth, 182 people were certified over 12,544 person-years. In the highest fifth, 182 people were certified over 13,247 person-years. The same number of cases: 14.5 per 1,000 person-years against 13.7.

Fifth	Riboflavin, mg/day	People	Cases	Person-years	Cases per 1,000 person-years	Hazard ratio, men / women
Lowest	0.68	834	182	12,544	14.5	1 / 1
Second	0.96	834	183	12,432	14.7	0.59 / 0.95
Third	1.16	835	177	12,806	13.8	0.66 / 0.68
Fourth	1.41	834	166	13,281	12.5	0.47 / 0.63

Fifth	Riboflavin, mg/day	People	Cases	Person-years	Cases per 1,000 person-years	Hazard ratio, men / women
Highest	1.81	834	182	13,247	13.7	0.55 / 0.51

Counts from the preprint's Tables 1 and 2, men and women added together. Hazard ratios are the authors', adjusted for age, community, smoking, drinking, and calories.

The halving comes from the adjustment. Adjusting for age, sex, and community alone turns 13.7 against 14.5 into 0.55 for men and 0.51 for women. The paper does not show the counts by community, so a reader cannot see what the adjustment did. The counts do not show a halving. The model does.

Kishida et al., preprint, Tables 1 and 2.

Claim 2. Riboflavin cuts the risk

The fault: the causal leap. Two things go together, and one is taken to cause the other.

Nobody was given riboflavin. People ate what they ate, and the people in the top fifth ate differently in every way. Men in the top fifth ate 290 grams of vegetables a day against 215 in the bottom fifth; women 348 against 243. They ate more fish, more protein, and less rice, and their blood pressure was lower. The model adjusted for smoking, drinking, and calories. It did not adjust for vegetables, fish, education, income, or exercise. Any of those could produce the association on its own. "Cut" is a causal verb, and the study cannot support it.

Kishida et al., preprint, Table 1 and Methods.

Claim 3. Riboflavin stood out

The fault: the Bernoulli fallacy. A result that clears a significance threshold is read as proof, without asking how many results were tried.

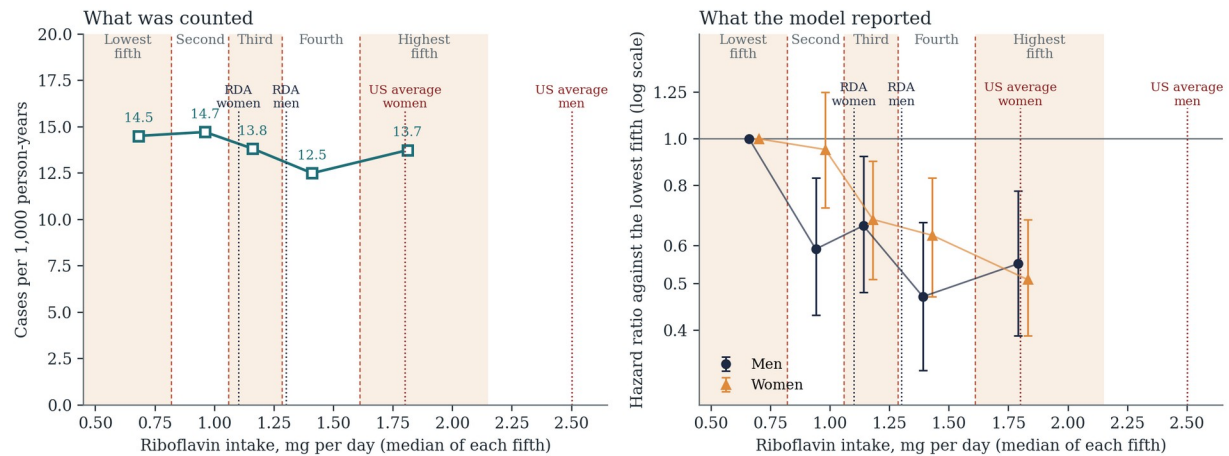
The study tested four vitamins, each in men and in women, each with and without a history of stroke: 24 comparisons. Riboflavin came out strongest. B12 mattered in men only, folate in women only, B6 in neither. The published paper then recut the data into quarters and pooled the sexes, and now B6 and folate show an effect and B12 does not. Riboflavin survived both cuts, which is a point in its favor. But the article reports the winner of 24 comparisons and leaves out the losers.

Kishida et al., preprint, Tables 2 and 3; published abstract.

The recommendation. Eat more riboflavin-rich foods

The fault: arbitrary categories. A continuous measure is cut into groups, and the cut points decide what the result says.

The bottom fifth ate a median of 0.68 mg a day, below the US recommended allowance of 1.1 mg for women and 1.3 mg for men. The top fifth, at 1.81 mg, is about where an average American already sits: US adults average 2.5 mg (men) and 1.8 mg (women) from food, mostly milk, bread, and fortified cereal. Fewer than 6 in 100 Americans fall below the allowance. The study compares people short of riboflavin with people who have enough. It says nothing about going from enough to more, and the recommendation is aimed at readers already past the study's top fifth.



Kishida R et al., Eur J Clin Nutr 2025;79:1099–1105. Counts and per-sex hazard ratios (adjusted for age, community, smoking, drinking, calories), 95% confidence bands, from the authors' preprint, Tables 1–2. Band edges drawn midway between the reported medians (illustrative). RDA and US average intakes: NIH Office of Dietary Supplements riboflavin fact sheet (NHANES 2009–2010).

The study's five fifths, drawn over the intake scale. Left, the counted rate of dementia certification: 14.5 per 1,000 person-years in the lowest fifth, 13.7 in the highest. Right, the model's hazard ratios with their 95% confidence bands. The bottom fifth sits below the US recommended allowance; the average American woman eats about what the top fifth ate, and the average American man eats more than anyone in the study.

The figure shows how far the claims outrun the data. On the left, the counted rate is nearly flat. The highest fifth is within one case per 1,000 person-years of the lowest. The halving in the headline lives on the right, in the model, and even there the confidence bands are wide and men and women disagree about which fifth did best. Then look at where the American reader stands. The whole study sits at or below the average American intake, and the average American man is off to the right of everyone in it. The recommendation asks readers to move along a stretch of the scale the study never measured.

What to ask

- How many people in each group got dementia, out of how many?
- What else was different about the people who ate the most riboflavin?
- How many vitamins, groups, and cuts were tried before this one was reported?
- Is my own intake inside the range the study measured?